

Sabeeli سَابِيلِي

How the AI works

Cited answers about Islam in your own language, from the approved sources only, and a da'i when it matters.

Open track · AI in the Service of Islamic Content Challenge 2026

Team: Mushari Alothman, Eman Saheli

Live: sabeeli-self.vercel.app · Code: github.com/mushari44/Islam-hackthon

A fluent answer is not a trustworthy answer

- Someone curious about Islam, or a new Muslim, asks a chatbot and gets fluent text with no source.
- Sometimes the verse is misquoted, or the hadith does not exist. They cannot tell which sentence to trust.
- A personal question (marriage, divorce, money) can get a personal "ruling" from a machine.

Sabeeli's promise

Every sentence shows its approved
source

Quran and hadith text is never
generated

Personal cases go to a person, not a
model

"Not found" beats a guess

One model, four narrow jobs

1 Analyse

Language, content level A-D, personal case, any quoted verse, and 2-4 search phrases in Arabic and English

2 Read a photo

Transcribes a photographed verse or saying. The check itself is done in code, not by the model

3 Write the answer

From retrieved passages only, citing every sentence. Scripture only as references

4 Draft the referral card

A summary for the da'i that the seeker edits and approves before anyone sees it

Model: Gemma 4 31B through OpenRouter (default), or Claude, behind one gateway (core/claude.py).

Only providers that neither store nor train on prompts are used. The same cited pipeline answers mentions of the Sabeeli assistant in community groups.

From question to cited answer



- **Not enough in the sources?** The reply says so and offers a da'i, instead of filling the gap.
- **No model available?** Sources-only mode shows the matching approved passages without a written answer.
- **Transparent:** every answer has a "How I found this" sheet with passages, scores, removals and timings.

Only the scholarly package's sources

Source	Package page	Records	Use
Mushaf text, At-Tafsir Al-Muyassar, Rowwad English (QuranEnc)	p. 9	6,236 verses	cited
HadeethEnc: text, grade, explanation, lessons	pp. 9-10	3,574 hadiths	cited
icadb Q&A encyclopedias (for non-Muslims, for Muslims)	p. 10	543 answers	cited
بينات: أسئلة وأجوبة عن الإسلام (built locally, rights reserved)	p. 4	263 questions	cited
Package glossary	p. 7	10 terms	cited
IslamHouse videos, streamed from IslamHouse	p. 9	107 languages	shown, never cited

Official APIs only, no scraping. Every record keeps its id and a link to the original page.

Verses quoted inside Q&A answers are re-linked to the Mushaf (1,849 markers), so the verse shown is always the Mushaf's.

Hybrid retrieval, measured

- **Natural units, never fixed chunks:** one verse, one hadith, one Q&A item, each with a stable id (q:2:256, h:2962).
- **Semantic:** multilingual E5-large with FAISS, run locally. Arabic and English with the same meaning land together.
- **Keyword:** Arabic BM25 with Uthmani-aware normalisation.
- **Fusion:** weighted reciprocal rank, 70% dense / 30% BM25.
- **Answer slot:** the closest approved Q&A answer ($E5 \geq 0.84$) always gets a place.
- **Abstain:** below the bar the reply is "not found", never a weak match.

Expected source among the 8 passages the model reads

17 / 23

BM25 alone

22 / 23

+ E5 hybrid

23 / 23

+ answer slot

The model cannot put words in the sources' mouth

Scripture by reference

The model writes `[[q:2:255]]` or `[[h:3062]]`. The app shows the stored verse or hadith. Any verse it types anyway is matched to the Mushaf and replaced.

Strict grounding

Every sentence must cite a retrieved passage or it is removed before display. Only short connectors pass (≤ 6 words, ≤ 3 at level D, no ruling words).

No citation, no answer

An answer left with no citation becomes the fixed "not found in the approved sources" message.

Invented quotes removed

A quoted saying found in no retrieved passage, such as a hadith from memory, removes its whole sentence.

Faithfulness check: median E5 similarity 0.84 between 287 cited sentences and their passages; all 34 under 0.80 were read by hand and were faithful.

Checking a quoted verse, word by word

- The seeker types or photographs a verse, or a saying presented as one.
- The model only transcribes the photo. The comparison with the Mushaf is deterministic code.
- Verdicts: exact · differs (each differing word shown) · several possible verses · not found in the Quran.
- A saying printed as if it were a verse is caught, with no model judgement involved.

Example

Typed by the seeker:

قل هو الله واحد

Mushaf (112:1):

قُلْ هُوَ اللَّهُ أَحَدٌ

Verdict: differs in one word

واحد بدل أحد

Scholarly safety by design

Level	What Sabeeli does
A	Answers with the source
B	Explains from approved material, with the reference
C	Notes that scholars differ, or refers to a specialist
D personal case	General information, the fatwa notice and a da'i. Never "allowed / required for you"

- **No personal fatwa** on any path, including sources-only mode.
- **Human in the loop:** a live call with a da'i who speaks the seeker's language.
- **Consent:** the referral card is shared only after the seeker approves it.
- **Privacy:** anonymous by default, questions deleted after 24 h, synthetic data only in tests.

Evaluated on 60 cases, 3 runs each

- 30 Arabic, 30 English, including all 12 sample cases from the package (p. 6).
- 20 quotes and photos · 20 explanations and follow-ups · 10 limits (missing source, invented hadith, disputed matter) · 10 personal fatwa.
- A run passes only if every check passes: behaviour, expected source, grounding, no generated scripture, no personal ruling.

Run	Passed
Baseline	88% (53/60)
After fixes	95% (57/60)
No model (fallback)	73% (44/60)
Final, 3 runs per case	100% (180/180)

Personal fatwa: 30 / 30 safe

Fast and cheap enough to run for real

2.8 s

median per answer
(3.7 s at p90)

76 ms

all of retrieval
(E5 26 · FAISS 27 · BM25 6)

\$0.0007

per question
with Gemma 4 31B

204

automated tests,
model mocked

The whole evaluation (300+ model answers) cost about \$0.22. Every answer carries per-step timings, and the slow outliers are OpenRouter hosts, not retrieval.

Deployment: website on Vercel, API on Render (Docker). Calls use WebRTC with optional TURN.

What we know is not done yet

- **Scholarly sign-off:** ask the reviewer to confirm At-Tafsir Al-Muyassar (p. 3 names other tafsir sources) and the corpus as a whole.
- **Live server:** the free host runs keyword search only and without بيانات, so it says "not found" more often than the evaluated setup. No check is loosened.
- **Short scripture:** a verse under 7 words typed by the model without quotation marks can slip past the guard. It never reached the screen in the evaluation.
- **English:** the Q&A sources are Arabic only; next, the package's English sources (icadb translations, IslamHouse, Byenah).
- **Next:** stream answers, and measure benefit with real seekers and da'is.